Achraf Azize

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achraf-azize.github.io

Current Position

2026–Present **Assistant Professor**, *Toulouse School of Economics*, Toulouse, France
I am an Assistant Professor in the Department of Mathematics and Statistics at Toulouse School of Economics (TSE), where I hold a Chaire de Professeur Junior (CPJ). My research explores the interplay between privacy, statistical learning, and economic incentives.

Education

2021–2024 **Ph.D. in Machine Learning**, *Inria Scool, Université de Lille*, France
Title: *Privacy-Utility Trade-offs in Sequential Decision-Making under Uncertainty*
Supervisors: Debabrota Basu, Philippe Preux. Link: HAL

2020–2021 **M.Sc. in Machine Learning**, *MVA, ENS Paris-Saclay*, France
Machine Learning, Applied Maths and Statistics

2017–2021 **Diplôme d'ingénieur**, *École Polytechnique*, France
Applied Maths, Computer Science and Physics

Research and Work Experience

Academic Positions

2025–2026 **Postdoctoral Researcher**, *Inria FairPlay, CREST, ENSAE*, France
Privacy and competitive analysis in online learning. Advisor: Vianney Perchet

2021–2024 **Graduate Researcher**, *Inria Scool, Université de Lille*, France
Differential Privacy, Multi-armed bandits, Privacy auditing, Membership Inference

Visiting Stays and Internships

2025 **Visiting Researcher**, *Boston University*, USA
Collaboration with Adam Smith on interactive Differential Privacy for contextual bandits

2024 **Visiting Researcher**, *Kyoto University*, Japan
Collaboration with Junya Honda on Differential Privacy for adversarial bandits

2021 **Research Intern**, *InstaDeep*, France
Multi-Object Manipulation using Reinforcement Learning and Graph Neural Networks.

2020 **Research Intern**, *DataLab Groupe Crédit Agricole*, France
Worked on AutoML pipelines and interpretability for real-world tabular data

Teaching

2026–Present **Teaching Assistant**, *L2 Eco/Math*, TSE, France
Probability tutorials.

2021–Present **Teaching Assistant**, *MVA, ENS Paris-Saclay*, France
Graphs in Machine Learning, with Daniele Calandriello. Course link

2022–2023 **Lecturer**, *École Centrale de Lille*, France
Python Practicals for the SDIA Masters. Course link

Publications

[1] **A. Azize** and D. Basu. When privacy meets partial information: A refined analysis of differentially private bandits. *NeurIPS*, 2022.

[2] **A. Azize** and D. Basu. Rényi differentially private bandits. *PPAI@AAAI*, 2023.

[3] **A. Azize**, M. Jourdan, A. Al Marjani, and D. Basu. On the complexity of differentially private best-arm identification with fixed confidence. *NeurIPS*, 2023.

[4] **A. Azize** and D. Basu. Concentrated differential privacy for bandits. *IEEE SaTML*, 2024.

[5] **A. Azize** and D. Basu. Open problem: What is the complexity of joint differential privacy in linear contextual bandits? *COLT*, 2024.

[6] **A. Azize** and D. Basu. Some targets are harder to identify than others: Quantifying the target-dependent membership leakage. *AISTATS*, 2025 (**Oral Presentation**).

[7] **A. Azize**, Y. Wu, J. Honda, F. Orabona, S. Ito, and D. Basu. Optimal regret of bandits under differential privacy. *NeurIPS*, 2025.

[8] M. Jourdan and **A. Azize**. Optimal best arm identification under differential privacy. *NeurIPS*, 2025.

[9] **A. Azize**, M. Jourdan, A. Al Marjani, and D. Basu. Differentially private best-arm identification. *JMLR, 2026*.

[10] G. Genalti, **A. Azize**, and V. Perchet. Online packet scheduling with deadlines and learning. *ICML, 2026*.

[11] **A. Azize**, M. Tamine, B. Heymann, I. Sebag, A. Betlei, C. Odic, M. Vono, P. Loiseau, and V. Perchet. Mirror descent for private learning with out-of-distribution public data. *(Under Review)*.

[12] **A. Azize**, M. Molina, H. Richard, and V. Perchet. Prophet inequalities under local differential privacy. *EC, 2026*.

Research Activities

Invited Talks and Posters

2026 **FairPlay Seminar**, *Criteo*, Paris
Telecom S2A Seminar, *Telecom*, Paris
TSE Seminar, *Toulouse*, France
Mostly Monte Carlo Seminar, *Parisanté Campus*, Paris
HeKa Team Seminar, *Parisanté Campus*, Paris

2025 **FairPlay Team Retreat**, *Fréjus*, France
Boston Privacy Seminar, *Boston University*, USA
METIS Spring School Invited Talk, *UM6P in Rabat*, Morocco
AISTATS Oral and Poster Presentations, *Splash Beach Resort in Mai Khao*, Thailand
FairPlay Seminar, *Criteo*, Paris

2024 **SaTML Talk and Poster Session**, *University of Toronto*, Canada
COLT, *University of Alberta*, Canada
Shimo Lab Seminar, *Kyoto University*, Japan
IBIS Poster Session, *Tokyo*, Japan

2023 **Scool Seminar**, *Inria Lille*, France
Comete Ethical AI Workshop, *LIX, École Polytechnique*, France
NeurIPS Poster Session, *New Orleans*, US
RLSS Poster Session, *Barcelona*, Spain
EWRL Poster Session, *Brussels*, Belgium

2022 **Scool Seminar**, *Inria Lille*, France
NeurIPS Poster Session, *New Orleans*, US
EWRL Poster Session, *Politecnico di Milano*, Italy
NeurIPS @ Paris Talk and Poster Session, *Paris*, France

Reviewing

NeurIPS (2023-2024-2025), ICML (2022, 2023, 2024, 2026), COLT (2025), EWRL (2022, 2023, 2026), AAAI (2023), AISTATS (2025-**Best Reviewer Award**, 2026, 2027), TMLR, IEEE T-IFS, JMLR, JRSSB, ITCS (2027)

Honors and Awards

- Holder of a Chaire de Professeur Junior (CPJ), 2026 - 2031
- AISTATS 2025 Best Reviewer Award and Oral Presentation
- Recipient of AI_PhD@Lille Fellowship, 2021 - 2024, THIA ANR program.
- French Government Excellence Scholarship, 2017-2021 (Top 7 in Morocco).
- Member of the Moroccan Mathematics Olympiad Team (Top 12 in Morocco).
- Ranked 2nd out of 1779 candidates in the Concours National Commun 2017 (CNC)

Skills

Languages	Arabic (native), French (fluent), English (fluent)
Coding	Python, SQL, Go, Java, C/C++
Tools	Git, Google Colab, Google Cloud Platform, PyTorch
Interests	Football, Tennis, Pétanque, Swimming, Music